Module 1
Introduction to RAG

Welcome to the Course
- Course Introduction — Video • 2 min
- Course Overview — Reading • 4 min
- RAG and Agentic AI Professional Certificate Overview — Video • 5 min
- Reading: Helpful Tips for Course Completion — Ungraded Plugin • 5 min

What is RAG?
- Why RAG? — Video • 6 min
- More RAG Details — Video • 7 min
- What is RAG? — Reading • 15 min
- Understanding the Why and How of RAG — Dialogue • 15 min
- Summarize Private Documents Using RAG, LangChain, and LLMs — Ungraded App Item • 45 min
- Building the Case: Explaining RAG Architecture to Your Product Stakeholder — Role Play • 15 min
- Practice Quiz: What is RAG? — Practice Assignment • 15 min

Summary and Evaluation
- Summary and Highlights: Introduction to RAG — Reading • 2 min
- Cheat Sheet: Introduction to RAG — Reading • 5 min
- Graded Quiz: Introduction to RAG — Graded Assignment • 30 min

Module 2
Build Apps with RAG

Module 3
Build RAG Apps with LlamaIndex

Course Overview

16 XP • Advanced RAG System Implementation | 3 XP • Resilient Systems and Third-Party Integrations

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Hello, and welcome to the **Build RAG Applications: Get Started** course.

Designed for aspiring artificial intelligence (AI) developers, machine learning engineers, data scientists, and AI researchers, this course provides the knowledge and skills to develop your skills with two popular retrieval-augmented generation (RAG) frameworks. Throughout this course, you'll gain hands-on experience with LangChain and LlamaIndex, two popular frameworks used for building in-demand AI-powered apps that use RAG.

You'll begin by exploring RAG fundamentals, learning how RAG enhances information retrieval and user interactions, before building your first RAG pipeline.

Next, you'll discover how to create user-friendly AI applications using Gradio, moving from project planning to constructing a QA bot using LangChain that can answer questions from loaded documents.

You'll compare two RAG frameworks, LlamaIndex and LangChain, and then develop an application with LlamaIndex to gain experience with a different RAG framework. Throughout the course, you'll engage in interactive labs and leverage multiple large language models (LLMs), ensuring you have the skills to design, implement, and deploy RAG applications.

This course is part of the **IBM RAG and Agentic AI Professional Certificate**, built to provide you with practical skills and knowledge you can use to excel in developing advanced AI applications that leverage RAG, multimodal AI, and agentic AI systems.

Prerequisites

This course requires working Python programming skills and experience, as the course emphasizes coding and application development. Additionally, fundamental knowledge of AI and web development is highly recommended. A strong foundation in generative AI is recommended but not required.

Objectives

After completing this course, you will be able to:

- Explain why developers use RAG to build AI-powered applications.
- Design user-friendly, interactive interfaces for RAG applications using Gradio.
- Describe how the LlamaIndex framework supports building RAG applications and how it contrasts with LangChain.
- Build RAG applications using LangChain and LlamaIndex.

Course Outline

This course consists of three modules.

Module 1: What is RAG

- Lesson 0: Welcome
- Lesson 1: What is RAG?
- Lesson 2: Module Summary and Evaluation

Key Topics:

- Why RAG
- More RAG details
- Summarize Private Documents using RAG, LangChain, and LLMs

Module 2: Build Apps with RAG

- Lesson 1: Create an Interactive RAG Application with a User-Friendly Gradio Interface
- Lesson 2: Module Summary and Evaluation

Key Topics:

- Getting Started with Gradio
- Introduction to Gradio
- Set Up a Simple Gradio Interface to Interact with Your Models
- Construct a QA Bot That Leverages the LangChain and LLM to Answer Questions from Loaded Documents

Module 3: Build RAG Apps with LlamaIndex

- Lesson 1: Application Development with LlamaIndex
- Lesson 2: Module Summary and Evaluation
- Lesson 3: Course Wrap-Up

Key Topics:

- Intro to LlamaIndex: From Vector Stores to Query Engines
- Intro to LlamaIndex: From Vector Stores to Query Engines
- LangChain and LlamaIndex Compared
- Build an AI Icebreaker Bot with IBM Granite 3.0 & andLlamaIndex

Tools/Software

In this course, you will explore and utilize a variety of tools and platforms, including:

- LangChain and LlamaIndex to design and implement structured AI workflows
- IBM Granite, a trusted source of AI models purpose-built for business and engineered to ensure trust and scalability in AI-driven applications
- Gradio, an open-source Python package for quickly building demos or web applications for machine learning models, APIs, or any arbitrary Python function
- Python for coding AI applications, integrating APIs, and implementing LangChain's functionalities effectively

Tips for Success

- Stay organized and track key program milestones, risks, and stakeholder communications
- Consistently practice using the generative AI tools and techniques covered in the course

Congratulations on taking this step to boost your career with the RAG application development skills. We hope you enjoy your learning journey.

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